

ARXIV 2504.05299

SmolVLM

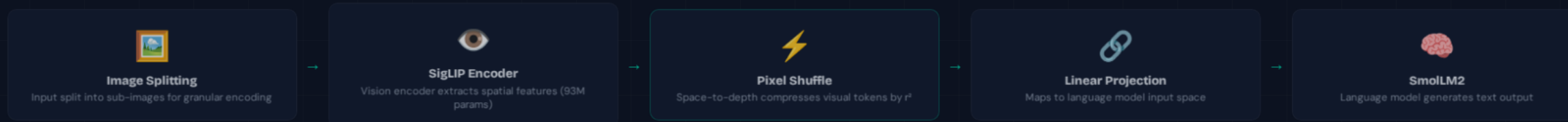
Redefining Small & Efficient Multimodal Models

Compact vision-language models achieving competitive performance at sub-1 GB RAM — from
Hugging Face & Stanford University



ARCHITECTURE

SmolVLM Pipeline: Image Splitting → Pixel Shuffle → SmolLM2



FINDING 1

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Config A — Balanced

✓ Better Performance

SigLIP-B/16 · 93M

SmoLLM2 · 135M

Vision Encoder: 93M

Language Model: 135M

Total: 228M

Config B — Oversized Encoder

✗ Worse Performance

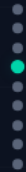
SigLIP-S0400M · 428M

SmoLLM2 · 135M

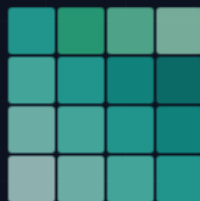
Vision Encoder: 428M

Language Model: 135M

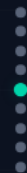
Total: 563M



FINDING 3

Aggressive Pixel Shuffle ($r=4$) Cuts Visual Tokens 16× $H \times W$ spatial tokensPixel Shuffle
 $r = 4$

16C

 $H/4 \times W/4 \times r^2 C$ $r = 2$ 4× reduction
Preferred for
larger models $r = 3$ 9× reduction
Mid-range
trade-off $r = 4$ 16× reduction
Best for small ViMs
— eases attention

FINDING 2

Extending Context from 2K to 16K Tokens Unlocks Higher Resolution

01

RoPE Base Adjustment

Increased from **10K** to **273K**,
enabling extrapolation to longer
sequences without positional collapse

02

Mixed-Length Finetuning

Trained on a blend of long & short-
context data to retain short-context
performance while gaining long-range
reasoning

03

Model-Specific Limits

2.2B model uses **16K** context;
smaller variants (**256M**, **500M**)
capped at **8K** tokens



BENCHMARKS

Competitive Performance at Sub-1 GB to 4.9 GB RAM

Category	Benchmark	256M	500M	2.2B	Best Efficient OS
Single-Image	OCRBench	52.6	61.0	72.9	54.7 (MoIms)
Single-Image	ChartQA	55.6	62.8	68.7	48.0 (MoIms)
Single-Image	TextVQA	50.2	60.2	73.0	61.5 (MoIms)
Single-Image	DocVQA	58.3	70.5	80.0	77.7 (MoIms)
Multi-task	MMMU	29.0	33.7	42.0	33.9 (MoIms)
Multi-task	MathVista	35.9	40.1	51.5	37.6 (MoIms)
Video	Video-MME	33.7	42.2	52.1	45.0 (InternVL2)
Video	MLVU	40.6	47.3	55.2	48.2 (InternVL2)
Average (13 benchmarks)		44.0%	51.0%	59.8%	



ON-DEVICE

Real-Time Inference on Apple Silicon: Up to 80 tok/s

~80

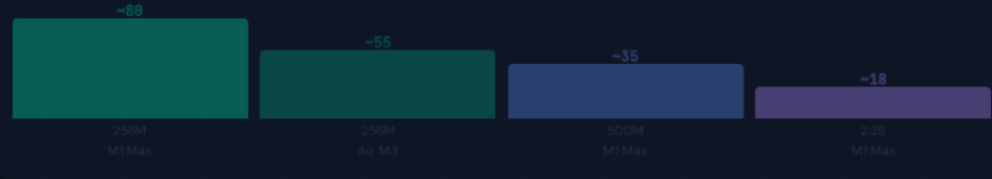
tokens / second
SmolVLM-256M on
M1 Max 64 GB (batch 1)

~1000

tokens / second
SmolVLM-256M at
batch 64 on high-end devices

~55

tokens / second
SmolVLM-256M on
MacBook Air M3 16 GB



TAKEAWAYS

Design Principles for Efficient Multimodal Models



Balance Encoder-LM Compute

Smaller vision encoders outperform larger ones when the language model is compact. Match encoder scale to LM capacity.



Extend Context Length

Scaling from 2K – 16K tokens via RoPE base tuning unlocks higher image resolution and multi-image reasoning.



Aggressive Token Compression

Pixel shuffle at $r=4$ yields 16 $\times$ token reduction — ideal for small VLMs where attention overhead dominates.



Sub-1 GB Inference Is Achievable

SmoIVLM-256M runs at 0.8 GB RAM, enabling deployment on smartphones and low-power edge devices.



Fully Open-Source Release

All weights, datasets, training code, and the HuggingSnap mobile app are open-sourced to advance on-device AI research and reproducibility.

